Perfect Competition and Cost

Answer key with worked solutions

Profit = TR - VC - FC. Variable profit = TR - VC. AC means ATC. Fixed costs remain payable during a short-run shutdown. Surplus questions use PS = TR - VC and TS = CS + PS.

Source: Perfect Competition and Cost (1).pdf

#	Answer	Work / explanation
1	**a, b**	In long-run competitive equilibrium, P = MR = MC = minimum ATC.
2	**d**	300 = 0.5q gives q = 600. TR = 300(600) = 180,000; VC = 0.25(600)² = 90,000. Profit = 180,000 - 90,000 - 100,000 = -$10,000. Continue operating because revenue covers variable costs.
3	**b, c, d**	900 = 0.5q gives q = 1,800. Profit = 900(1,800) - 0.25(1,800)² - 100,000 = $710,000. Entry lowers price. Long-run MC = ATC: 0.5q = 100,000/q + 0.25q, giving q ≈ 632.46 and P ≈ $316.23. This equals the new MC and is below the current MC of $900.
4	**a**	Positive profit requires P > AC. Since P = MC, MC > AC.
5	**b**	The owner’s forgone earnings are an implicit opportunity cost, deducted when calculating economic profit.
6	**D - $10**	MC = 2q gives q = P/2. Market supply: Qs = 10(P/2) = 5P. Equilibrium: 5P = 60 - P, so P = $10 and Q = 50.
7	**e - $1**	Minimum AVC is $10. At $1, the firm shuts down and earns profit = -FC. All other listed prices generate a smaller loss or a positive profit.
8	**e - None DWL = $0**	The competitive equilibrium equates marginal benefit and marginal cost. With the given firms, deadweight loss is $0.
9	**c**	Profit per unit = 0.50 - 0.75 = -$0.25. Firms exit, so market supply decreases and price increases.
10	**a, b, c, e**	20 = 2q gives q = 10. TR = $200; VC = q² = $100. Variable profit = $100. Total profit = 200 - 100 - 2,000 = -$1,900. Continue operating and produce 10 units.
11	**c, d**	Forgone return = 100,000(0.05) = $5,000. This opportunity cost is deducted when calculating economic profit.
12	**c**	Profit per unit = 6 - 9 = -$3. Exit causes market supply to decrease and price to increase.
13	**a - $15**	The graph gives q = 5 and AC = $7 at P = $10. Profit = (P - AC)q = (10 - 7)(5) = $15.

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14	**d - Less than $7**	Long-run price equals minimum AC. The graph's AC curve reaches its minimum below $7.
15	**a, b, c, d, e (short run)**	3 = 0.1q gives q = 30. TR = $90; VC = 0.05(30)² = $45. Variable profit = $45 > 0, so operating reduces losses under every listed fixed cost. The question needs a time horizon: for a long-run decision at P = $3, only a and b cover total costs.
16	**B - $1,280**	Each firm supplies q = 0.5(8) = 4. Market quantity Q = 80(4) = 320. PS = ½(320)(8) = $1,280.
17	**c**	From the graph, MC ≈ 2 + 0.1q, so P = $5 gives q ≈ 30. TR ≈ $150; VC ≈ 2(30) + ½(30)(3) = $105. Variable profit is positive, but P < AC, so economic profit is negative. Exit causes price to rise.
18	**a - 400, 5**	In C(q) = 400 + 5q, fixed cost is $400 and each additional unit adds $5 to cost. Thus MC = $5.
19	**a, c, d, e**	Minimum AC is approximately $9: below $15 and above $6. Long-run price occurs where MC = AC, and economic profit is zero. The graph does not establish that this particular firm must exit.
20	**a - $1,250**	MC = 2q gives Qs = 5P. Equilibrium: 5P = 60 - P, so P = $10 and Q = 50. CS = ½(60 - 10)(50) = $1,250.
21	**a, b**	Below P_1 = minimum AVC, the firm shuts down but loses its fixed costs. Between P_1 and P_2 = minimum ATC, it operates with negative total profit.
22	**b - $2,880**	MC = 6q gives q = P/6. Qs = 60(P/6) = 10P. Equilibrium: 10P = 1,200 - 40P, so P = $24 and Q = 240. PS = ½(240)(24) = $2,880.
23	**a, b, c, e**	20 = 5q gives q = 4. TR = $80; VC = 2.5(4)² = $40. Variable profit = $40. Total profit = 80 - 40 - 100 = -$60. Continue operating and produce four units.
24	**b - 12 units**	The graph shows P = MC = $12 at q = 12 on the rising portion of MC.
25	**c - $9**	Long-run price equals minimum ATC, which the graph identifies as $9.
26	**a - $1,500**	At P = $10 and Q = 50: CS = ½(60 - 10)(50) = $1,250; PS = ½(10)(50) = $250. Total surplus = CS + PS = $1,500.
27	**Setup only**	MC = 2q gives q = P/2. Four firms give Qs = 2P. Equilibrium: 2P = 30 - P, so P = $10, Q = 20, and q = 5.
28	**a - $300**	CS = ½(30 - 10)(20) = $200; PS = ½(10)(20) = $100. Total surplus = 200 + 100 = $300.
29	**c - Four firms**	Each firm earns profit = 10(5) - [25 + 5²] = $0. There is no incentive for entry or exit.

#	Answer	Work / explanation
30	**b - No**	Operating profit = 10 - 50 = -$40. Shutdown profit = -$50. Operating reduces the loss by $10.
31	**c**	Same as Question 17. At P = $5, the firm covers variable costs but fails to cover total costs. Long-run exit reduces supply and raises price.
32	**a - $1,250**	Qs = 5P. Equilibrium: 5P = 60 - P, so P = $10 and Q = 50. CS = ½(60 - 10)(50) = $1,250. Fixed costs do not change short-run marginal cost or supply.
33	**e - None**	P = AC = $1 gives profit = $0. There is no entry or exit incentive: supply and price remain unchanged.
34	**E - None** **P ≈ $33.33**	MC = 4q gives q = P/4. Qs = 8(P/4) = 2P. Equilibrium: 2P = 100 - P, so P = 100/3 ≈ $33.33.
35	**c, d**	Forgone return = 50,000(0.10) = $5,000. This opportunity cost is deducted when calculating economic profit.